MEI LIUXI

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Professional Profile

Machine Learning researcher and Senior Software Engineer with a Master's in Statistics and Machine Learning. Combines over 5 years of industry experience in processing large-scale data and building distributed pipelines with a recent academic focus on applying **Graph Neural Networks (GNNs)** to **computational biology** and **omics data**. Interested in turning complex biological datasets into reliable models.

Work Experience

Senior Developer, SAP (System Applications and Products) *12/2021 – 08/2023*

- Created an Innovation Management System, and introduced it to collaborative enterprises such as Ericsson and Budweiser to help them identify the significant advice and feedback from internal and external consumers while improving the production process.
- Based on the natural language processing, extracted and categorized the core information and special requirements from the comments left on the internal requirements system of the company to enhance the improvement of the production.
- Was responsible for the modules of access control and role control for the innovation management system.

Senior Developer, NavInfo *09/2019 – 12/2021*

- Developed the distributed microservices architecture for the entire system.
- Combined DBSCAN and Graph matching algorithm to match the massive reported data of traffic element with the real traffic element.
- Was responsible for coding the module of stream process and the massive data from the terminal, integrated the algorithm to change the image data to the vector data.
- Analyzed the massive vector data post-back from the terminal on the car, and excavated the existing varied traffic features compared with the real digital map such as signs of messages and gas stations.
- Based on the initial analysis of all the traffic features, designed the enhanced Bayes algorithm to calculate the existing rate of the traffic features.
- Generated knowledge graph of the traffic elements based on the result of analysis and show the inference and decision process visually.

Developer, Ericsson Communication Technology Co., Ltd *07/2015 – 09/2019*

- Redesigned the part of 5G features to apply for 4G, strengthened the functions of 4G, and deployed it to the domestic test network.

- Analyzed the massive users' data recorded on the node, and utilized the unsupervised algorithm to discover the inner running pattern of the network of nodes.
- Designed the appropriate strategy and deep learning algorithm to minimize the electric bill of the network.
- Led the team to design and code the 5G power-saving platform, including the core algorithm, security strategy, and recovery strategy.
- Established a platform that reduced the electric bill for the telecommunication operators by 13%, the whole system worked for six provinces in China, controlled thousands of 5G nodes which served almost 100 million people.

Additional Information

- **Language Proficiency:** Mandarin (fluent), English (high proficiency).
- **Machine Learning & Bioinformatics:** Deep Learning, Graph Neural Networks (GNNs), Pathway and Network Analysis, Transciptomics/Omics Data Analysis, PyTorch Geometric, Scikit-learn.
- **Software Engineering & Big Data:** Proficient in Java, Python, R, TensorFlow, PyTorch, MySQL, Redis, Kafka, HBase, Distributed Microservices Architecture.

Education

Linköping University, Sweden *08/2024 – 06/2026*
Master's Degree, major in Statistics and Machine Learning

- **Master Thesis:** *Biologically informed graph neural networks for perturbation effect prediction*
- Proposed and developed a Counterfactual Attention GNN to predict drug-induced gene expression changes (omics data) by fusing molecular interaction networks with drug target annotations.
- Demonstrated that integrating biological topology and network analysis significantly improves model generalization on unseen drugs and cell lines compared to standard baselines.
- Extracted and analyzed graph attention weights to provide biological interpretability for the predicted molecular signaling pathways and to facilitate biomarker discovery.

Hubei University of Technology *09/2009 – 06/2013*
Bachelor of Engineering, major in Automation